

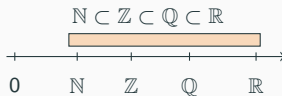
Math Bootcamp

Day 1 — Section 1.1: Notation and Number Systems

Math Camp 2026

1.1 Number systems — names and definitions

Symbol	English name	What it is
$\mathbb{N}$	natural numbers	Non-negative integers for counting
$\mathbb{Z}$	integers	$\dots, -2, -1, 0, 1, 2, \dots$
$\mathbb{Q}$	rational numbers	Fractions p/q ($q \neq 0$)
$\mathbb{R}$	real numbers	All points on the number line
$\mathbb{R}^n$	n -dimensional real space	Lists / vectors of length n



1.1 Symbols and vectors — names and definitions

Common symbols	$\in$	“is an element of”	$\subset$	“is a subset of”
	$\forall$	“for all”	$\exists$	“there exists”
	$\Rightarrow$	“implies”	$\Leftrightarrow$	“if and only if” (iff)

Scalar vs. vector

- **Scalar:** one number, e.g. income $w \in \mathbb{R}$
- **Vector:** a list of numbers $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$
- **Dot product:** $\mathbf{p} \cdot \mathbf{x} = \sum_{i=1}^n p_i x_i$

1.1 Logic — names and definitions

English	Symbol	Meaning
P only if Q	$P \Rightarrow Q$	Q is necessary for P (without Q , no P)
P if Q	$Q \Rightarrow P$	Q is sufficient for P (Q alone is enough)
P if and only if Q	$P \Leftrightarrow Q$	Q is necessary and sufficient
Contrapositive	$\neg Q \Rightarrow \neg P$	same truth as $P \Rightarrow Q$: if Q fails, P must fail

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Smallest set that fits?

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→ $\mathbb{N}$

counting: $0, 1, 2, \dots$

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e.g. $8.50 = 17/2$ dollars

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3. Order as a vector: (HK\$ spent on drink, # snacks) = (8.50, 2)

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→ $(8.50, 2) \in \mathbb{Q} \times \mathbb{N}$

1.1 — IKEA run Together

First week in HK: you go to IKEA and spot three plush toys.



Otter HK\$179.9



Hedgehog HK\$79.9



BLÅHAJ HK\$179.9

You have **HK\$500** pocket money ($B = 500$).

Basket $\mathbf{x} \in \mathbb{N}^3$: # otters, # hedgehogs, # sharks. E.g. $(1, 1, 0)$ = one otter, one hedgehog, no shark.

Price vector $\mathbf{p} = (179.9, 79.9, 179.9) \in \mathbb{Q}^3$. Budget set: $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$.

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$\mathbf{p} = (179.9, 79.9, 179.9)$, $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$. True or false?

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7. [contrapositive] **If** $\mathbf{p} \cdot (1, 1, 1) + \mathbf{p} \cdot (0, 1, 0) > 500$, **then** $(1, 1, 1) \notin \mathcal{B}$ **or** $(0, 1, 0) \notin \mathcal{B}$

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each affordable $\nRightarrow$ jointly affordable